Regional map refinement: problem, investigation and research

Prepared 2026-10-31 for review by a non-coding agent. Images are in `images/` next to this file. Captions that quote the product owner are marked "Owner"; everything else is the writer's description.

The report has three independent parts:

1. **Problem statement.** What the product owner asked for and what they saw, mostly in their own words.
2. **Investigation.** How the current system works and why it produces what the owner sees. It covers the work of two coding agents: Codex (Oct 29–31) and Claude (Oct 31).
3. **Research.** Which established techniques fit the problem, from terrain-synthesis research, geomorphology and cartographic generalization, and how they could be put together.

Each part can be read without the others. The investigation's conclusions do not depend on the research section.

Part 1 — Problem statement

1.1 The product

Vellum & Vector is a browser atlas for building fictional worlds used in tabletop roleplaying games (TTRPGs). It first generates a whole world: plate tectonics, climate, hydrology, biomes, peoples, history and magic, simulated on a mesh of about 60,000 cells. The default world is 4,000 × 3,000 km, so each world cell is roughly 14 km across.

A user can then select a **region** (about a twelfth of the world's land) and **refine** it. That creates a separate child map, sampled at about 2.4 km. A region can be refined again, with each level dividing into roughly 8–12 sub-regions.

Owner, on what the product is for:

> "the generation product is only there to create an intresting map to work on. it is not the main purpsose. there gis and numerical layers are all there to create world history. once the history is created it does not need to be coherenrt udner revision. the ENTIRE purpose of this product is to generate maps that would be used in a TTRPG where shit like numerical accuracy of elevation models does not matter"

On the relation between parent and child maps:

> "i expect the smaller refined maps to be fully seperate map sharing a coridnate system and world history, not that the refinment of a region would effect the larger world."

> "i DO NOT WANT there to be preservation of elevation across regions."

> "the land does not need to match between seperate regional refinments"

On what refinement should do:

> "what i expect to happen is that after selecting a region to edit refinment is done at the new selected scale (e.g. on a stnadard world squares were orignally 200kmsq, now they are at ~1.5kmsq. things should be deformed and fractalized acording to pysical processes. we can extract the physical motivation into reusuable things, but we should not bre repated history and everything else"

> "real refinment would produce geoegraphically distinct items, not just make every edge noiser."

1.2 The two competing goals (Owner, latest message, verbatim)

> "there are two competing goals of the system
>
> 1. maps should look like the maps at all levels. real maps of the world don't go down into micro details. autors take liberties with boundaries in order to draw a clear map. this isn't the sataline view, it's the artist view. regional maps have more details not just because they have more noise, but because more details are important. e.g. the missiipie and colorodo river show up on a world map, but the verde river doesn't show up until you look at a arizona map. lynx lake in prescott would only show up on a city map.
> 2. we need a complex geologic and numerical basis for these maps so there is authentistiy to them, and we cannot generate that at the level of detail required for the entire world. further, one of the eventual goals of this is collaberative map making, so diffrent people could own diffrent regions."

> "you seem unable to really get itme 1 right. you are to fixated on the numbers"

Restated as requirements (the writer's restatement, to be checked against the quotes above):

- **R1. Cartographic map, not satellite image.** Each map level shows the features that matter *at that level's scale*. A feature appears on a map because of its importance relative to the map's extent, not because it passes a fixed size threshold. Boundaries and shapes can be simplified or exaggerated for clarity.
- **R2. New detail is new features, not more noise.** A regional map of a mountain range shows named peaks, valleys, passes and sub-chains that the world map summarized as "a range". The analogy: the Rockies on a world map; Mt. Elbert, Mt. Massive and Mt. Harvard on a Colorado map.
- **R3. Physical authenticity.** Features have geological and hydrological causes. A lake sits where there is a reason for a lake. Rivers follow the terrain. Towns sit where people would settle.
- **R4. Detail is generated on demand.** The world simulation cannot run at fine resolution everywhere. Fine detail is produced only for the region being refined.
- **R5. Regions are independent maps.** Parent and child share a coordinate system and world history. Nothing propagates back to the parent. Parent elevations need not be preserved.
- **R6. Collaboration (future).** Different people may own and refine different regions. This implies each region's refinement should depend on the shared parent and the region's own seed, not on a neighbour's refinement.
- **R7. Performance.** World generation and region creation/boundary finding are fast today and should stay fast.

1.3 What the owner has observed

Figures 6–17 come from builds made by Codex. Figures 1–5 come from Claude's replacement, commit `bcca3a6`.

Latest build (Claude, `bcca3a6`)

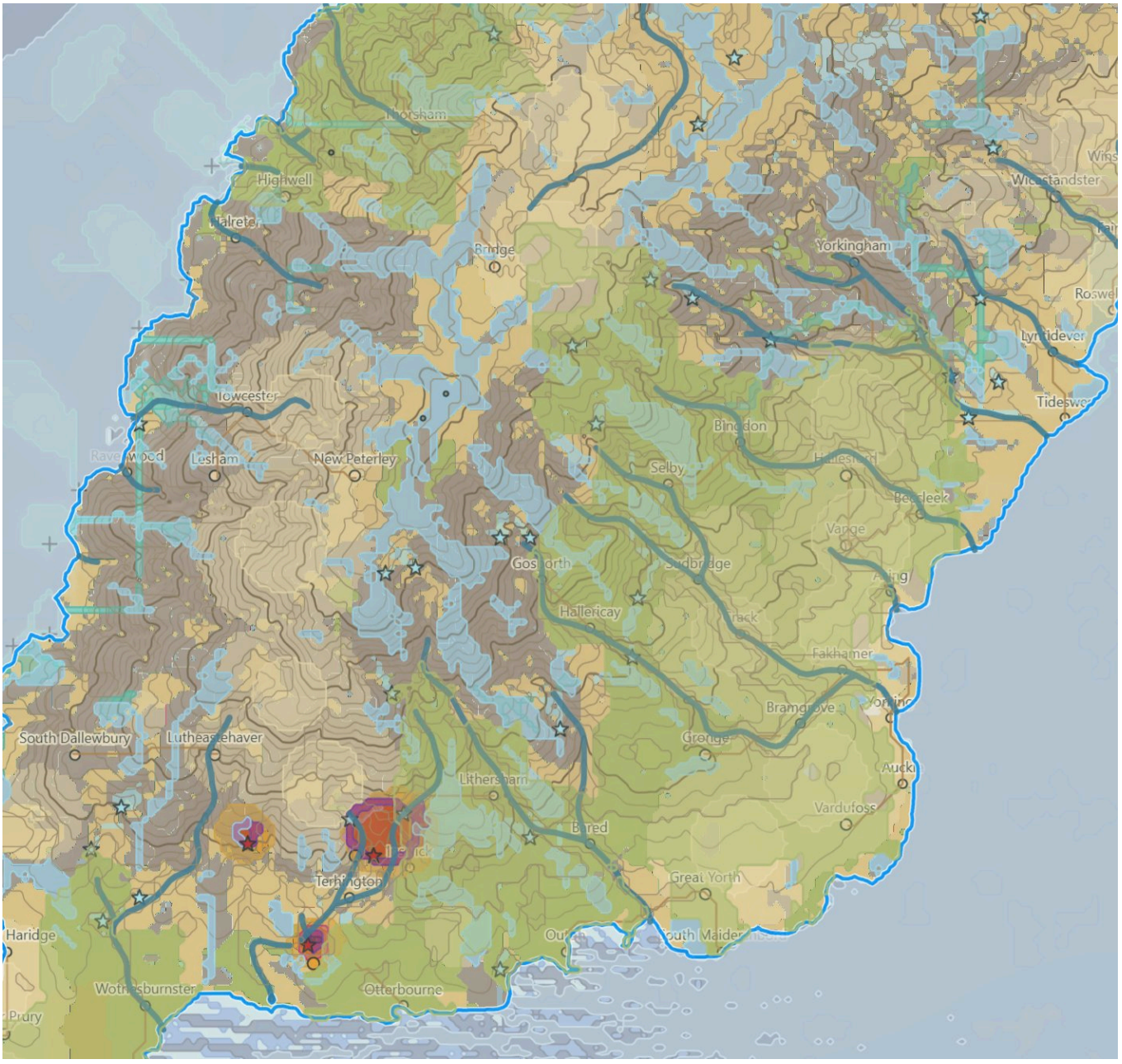


Figure 1. Latest regional map with the habitat layer. Owner: "thjings are 'better'".

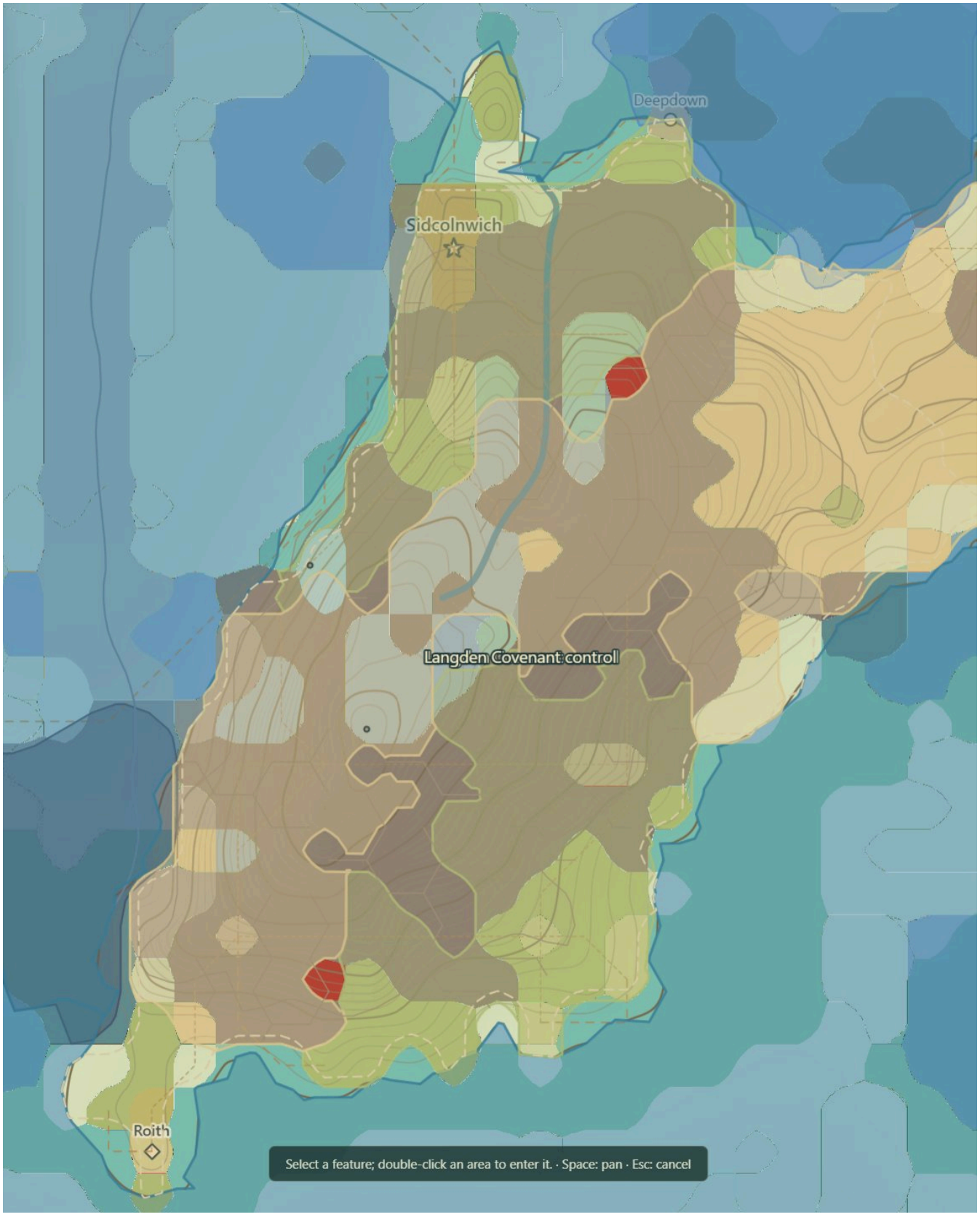


Figure 2. Second image the owner sent with the comment "thjings are 'better'". It appears to show a smaller landmass at a coarser level (writer's reading; not confirmed).

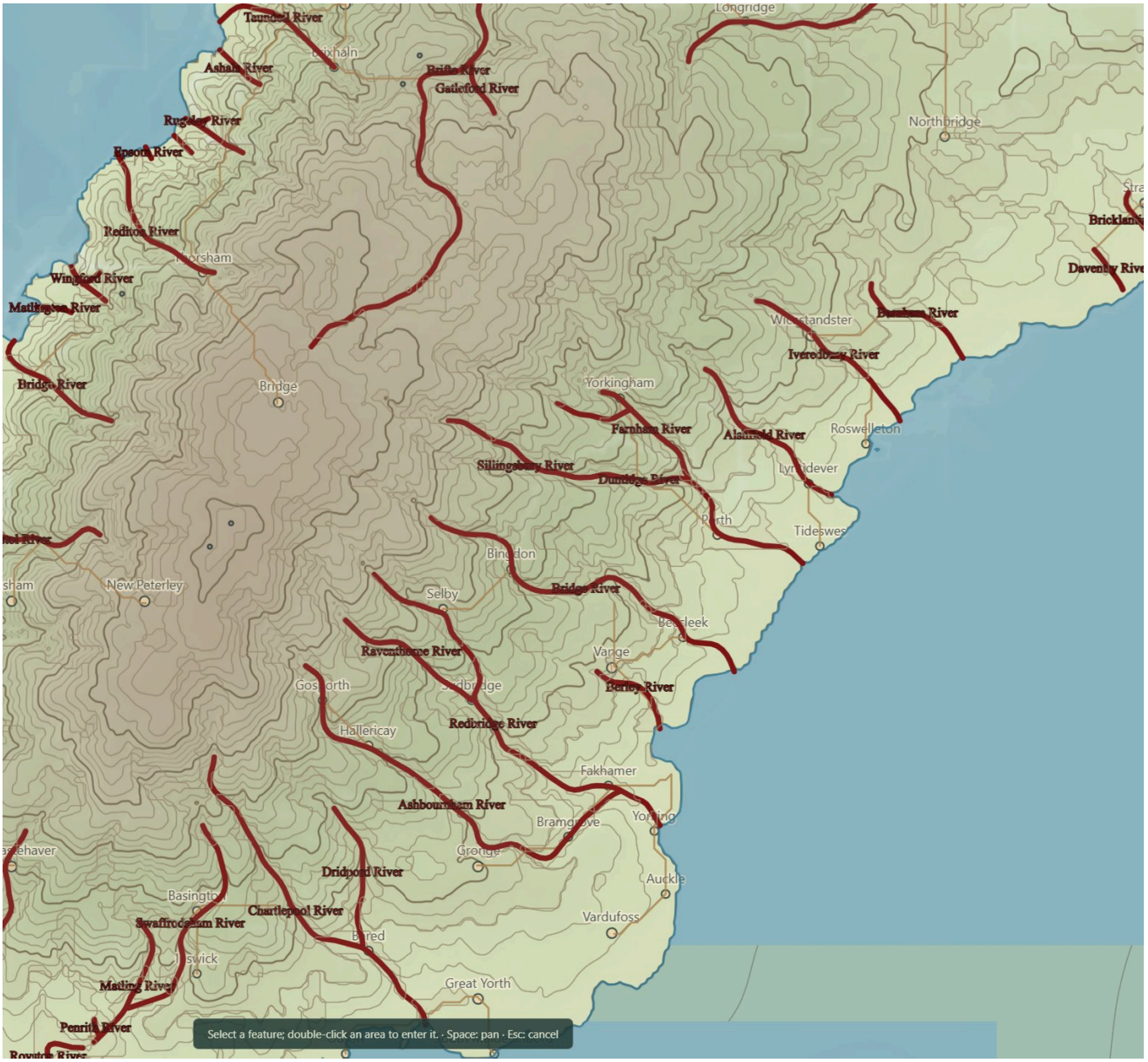


Figure 3. Latest regional map with rivers in red and elevation shading. Owner:

> "rivers are way to even, cities are better ish … although many cities are built into very steap cliff sides. there is no headland lakes. the elevation is till just noise - the regional map her represents a mountain range. it more or less looks appropraite for a world scale map, at world scale you see the rockies, not like elbert, massive, havard. but at the regional level major peaks should start to appear, and this is just one giant ridgeline"



Figure 4. Claude's own test render of a generated region, relief only. Each world-scale high area becomes one dome, with a comb of nearly parallel gullies running straight down every flank. There are no separate summits, cols or cross-valleys.

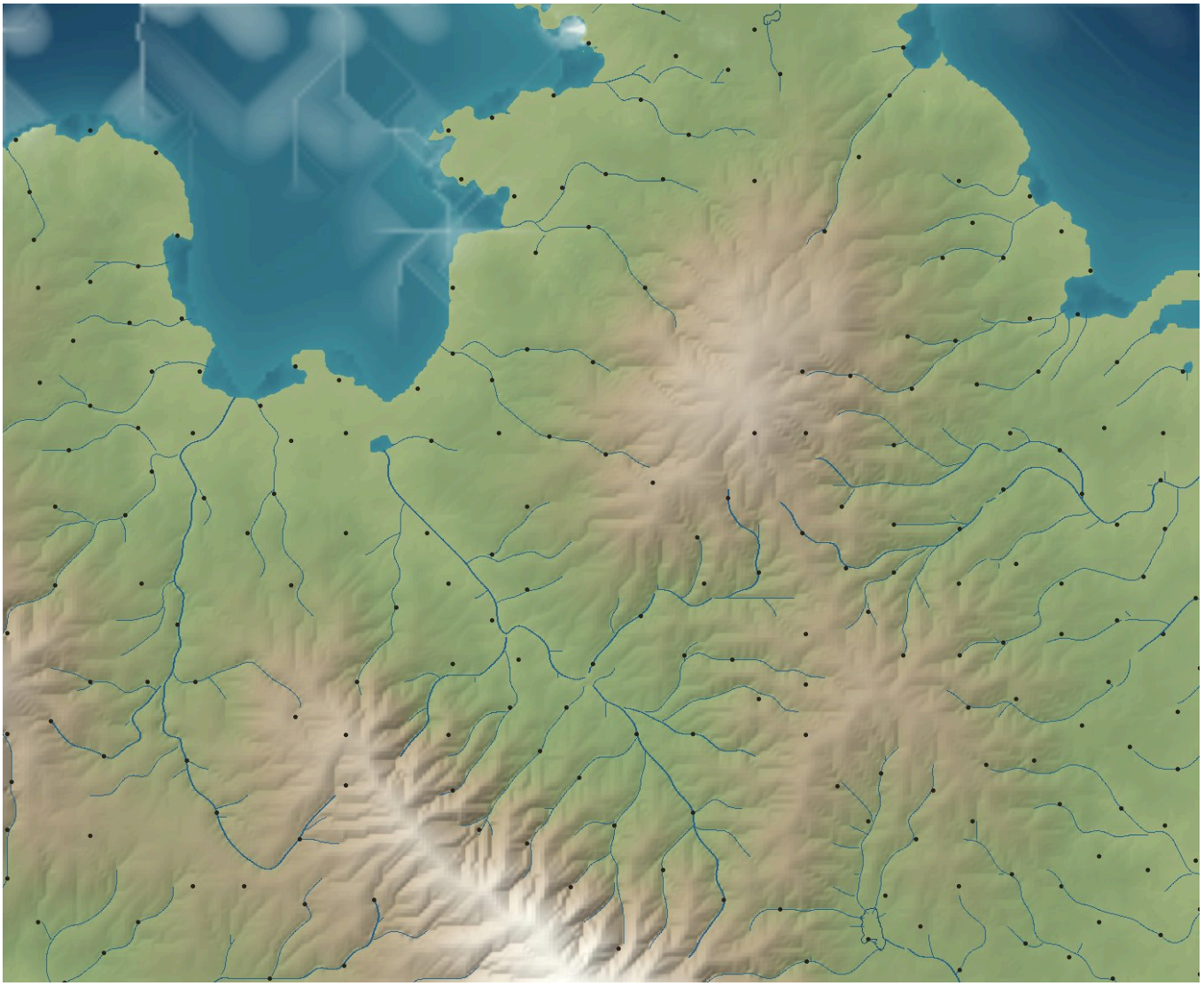


Figure 5. The same region with rivers and settlements. The rivers are a series of similar, evenly spaced streams.

Earlier builds (Codex)

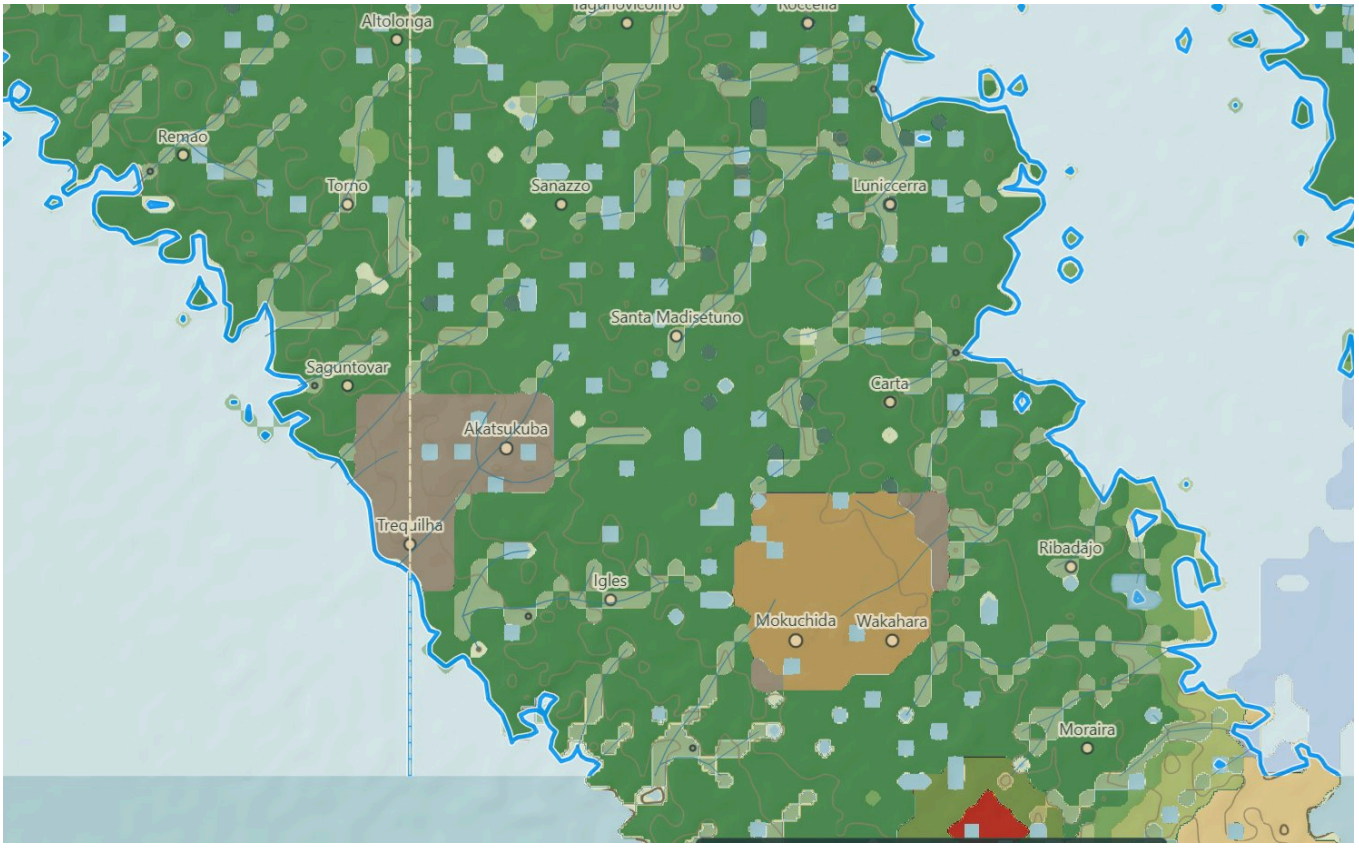


Figure 6. Owner: "ranges look more spiky, but they do not look more realistic. there are not more individual peaks. there are not valleys and passes and chains of mountsin. the existing toplogies just look noiser. … this isn't refiniemtn, it's just adding a lyer of noise."

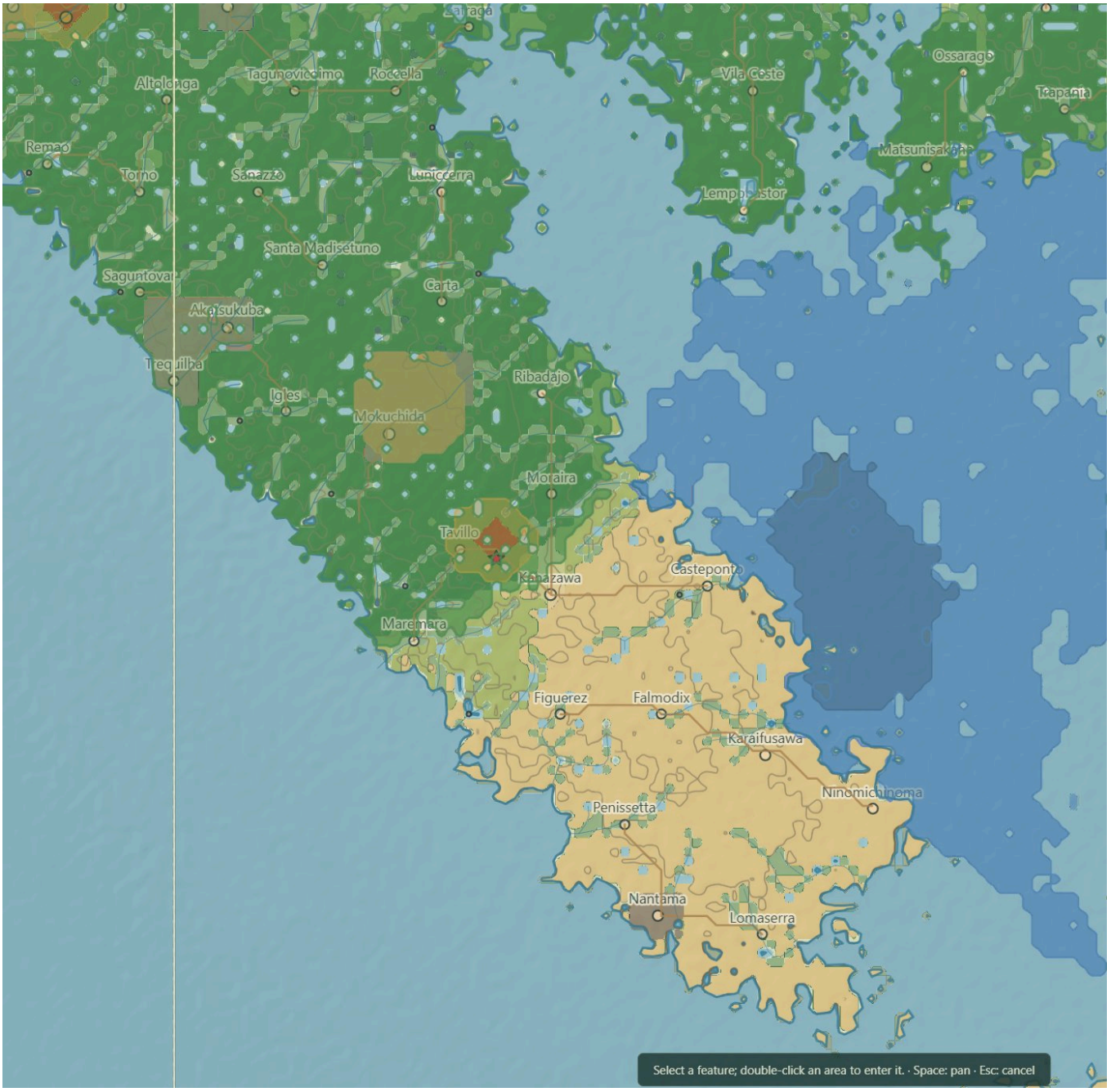


Figure 7. Owner: "towns are marginall better, but they seem to be spaced equidstnaatly, not on geographically relevant features."



Figure 8. Owner: "there are still a fuckload of lakes everywhere"



Figure 9. Owner: "i increased the line width of rivers to make them visible - there are a shitload on this map still"

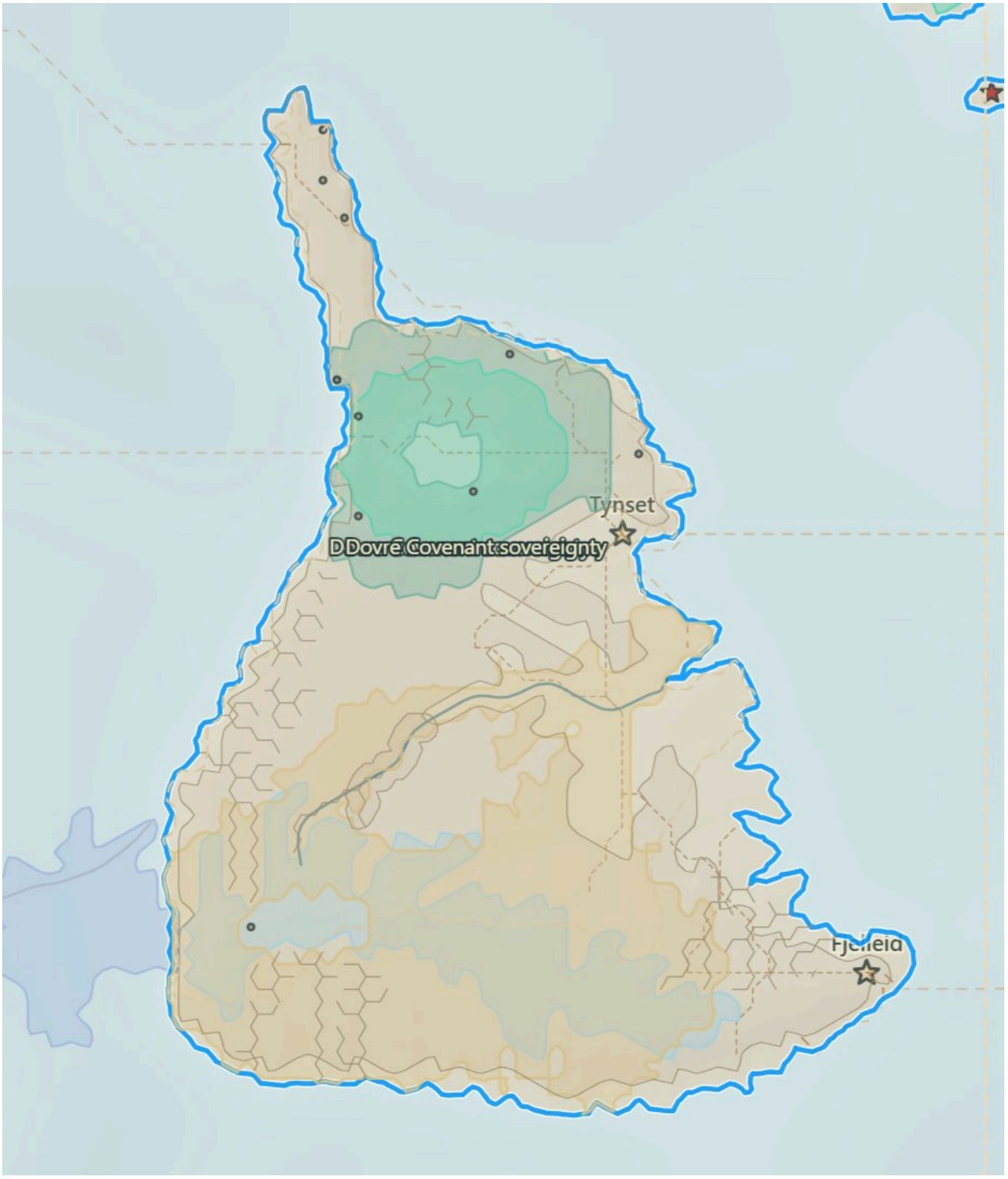


Figure 10. The parent (world) map of a refined region. Owner: "originally this was lableld as a range and rock barrens, with a few loose rivers and a larger one kindda in the middle. what i would expect to see happen here is maybe some larger lakes pop up - sources of the rivers, damns, whatever. but not a bunch of small ones. that's not really what a rock barren is"



Figure 11. Owner: "similar over on this other landmass"

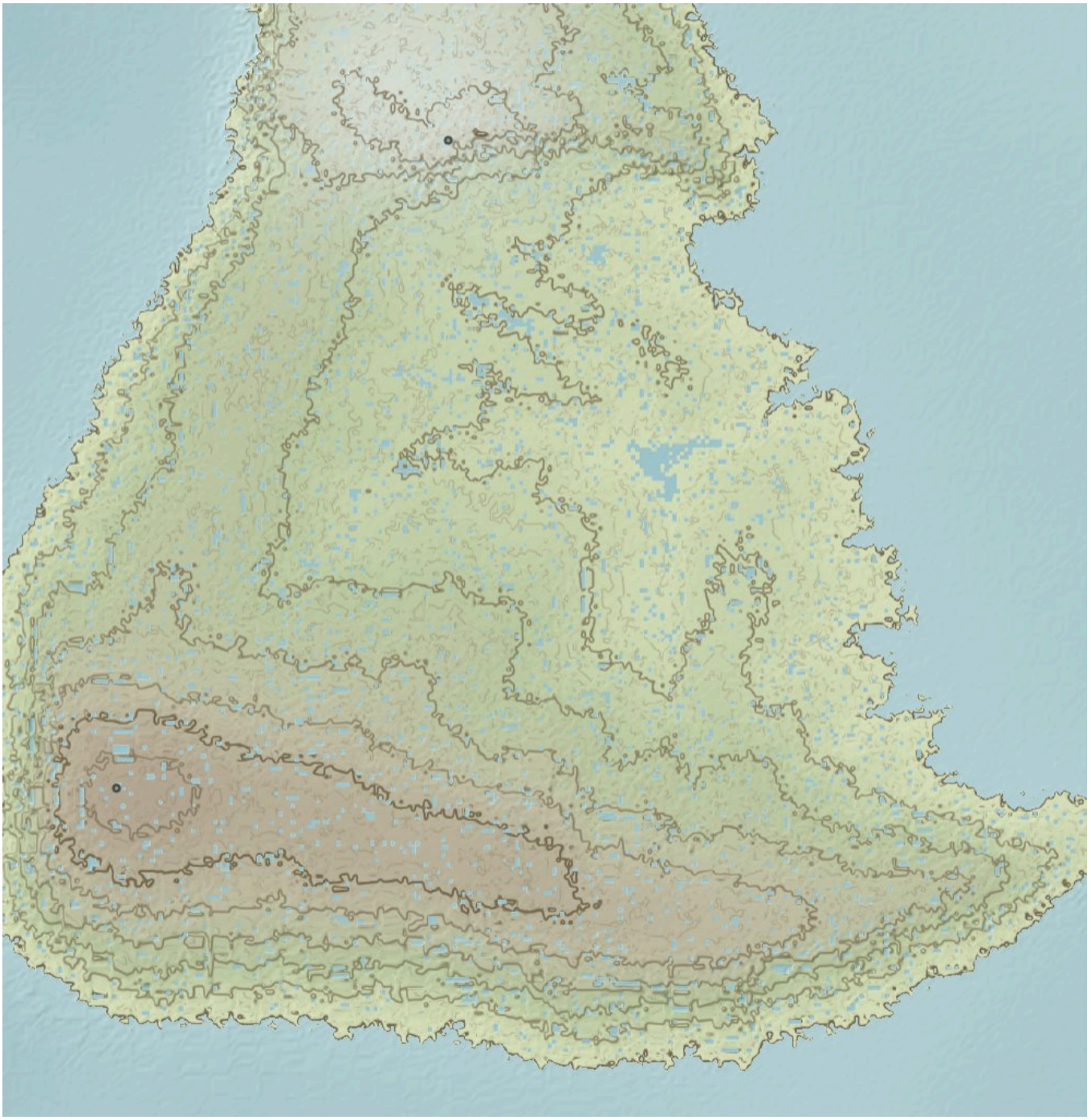
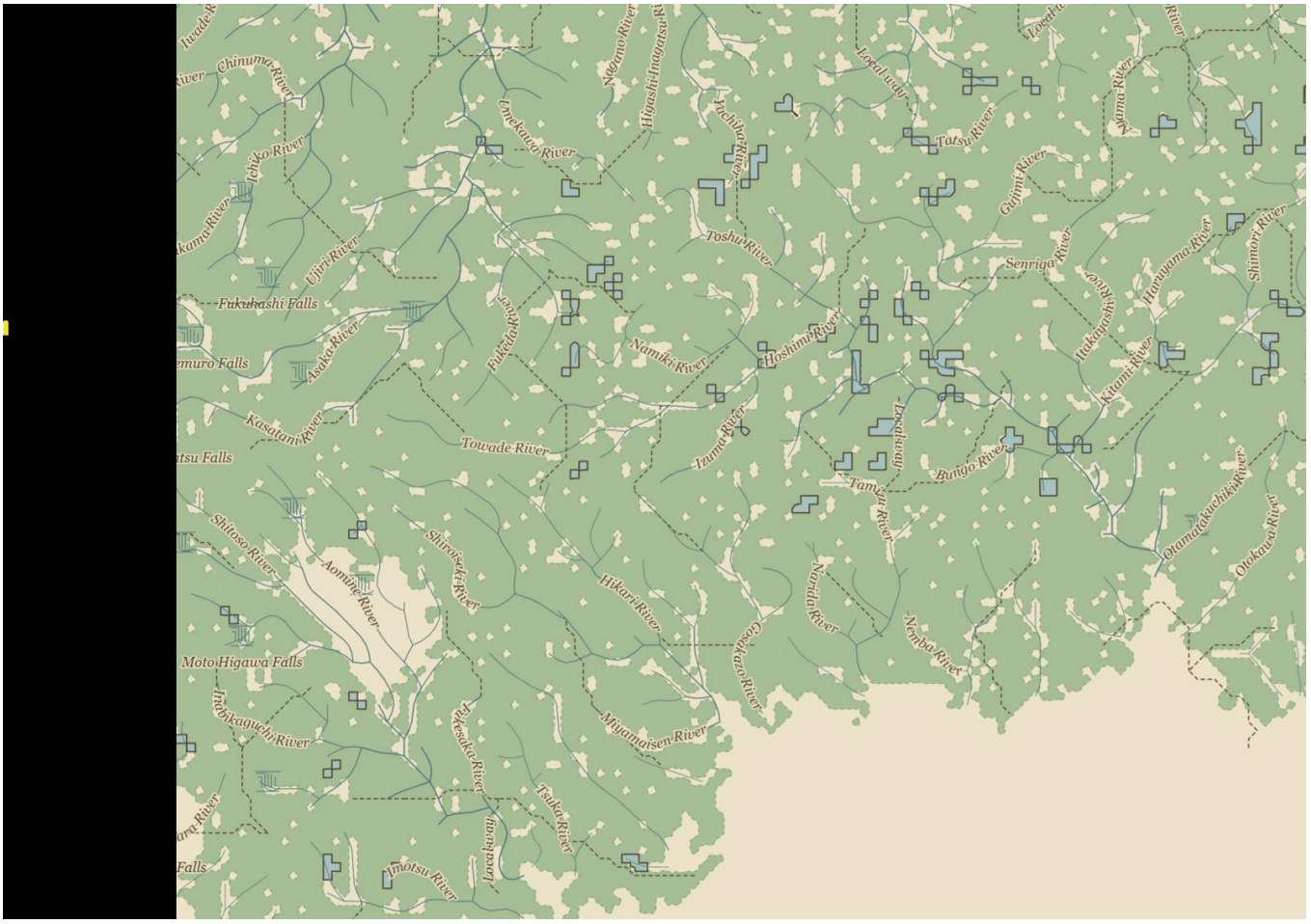
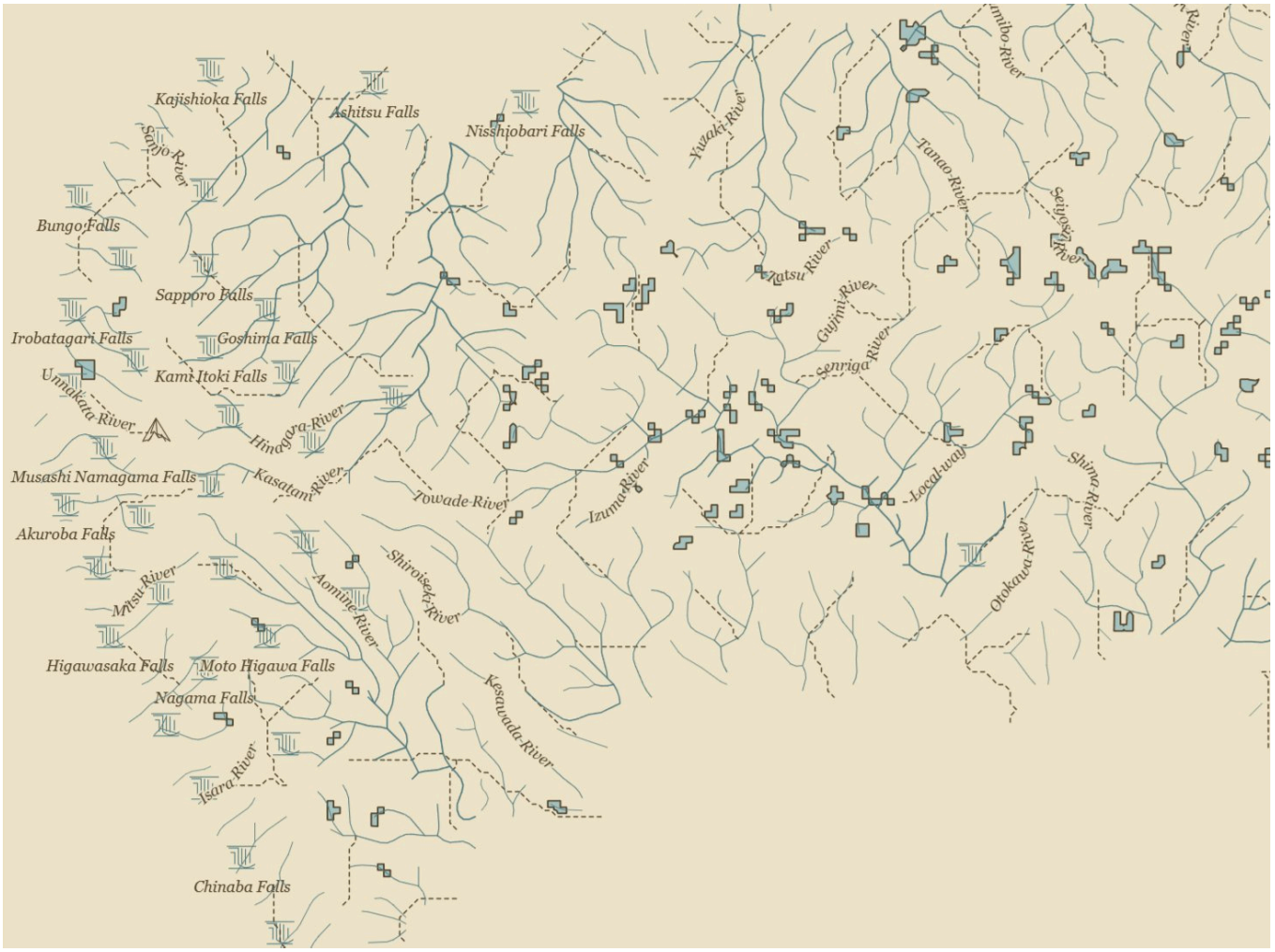


Figure 12. Owner: "elevation is not fixed". Speckled pits and lakes cover the land, and the contours are ragged.


Chinuma River
Ichiko River
Ujiri River
Fukuhashi Falls
Asaka River
Kasatani River
Towade River
Shitoso River
Aomine River
Shiroiseki River
Moto Higawa Falls
Hikari River
Miyamaisen River
Tsuke River
Toshu River
Namiki River
Hoshimi River
Izumi River
Tatsu River
Gujimi River
Senriga River
Kitami River
Bungo River
Tamizu River
Nemba River
Otamagakuchiki River
Otokawa River
Shimori River
Hotsumi River



Figures 13–14. Owner: "the gaps here are less obviously lakes … in this view they look more like holes in the forest. there are still for sure too many rivers for this zoom level. and the patchiness is not good."

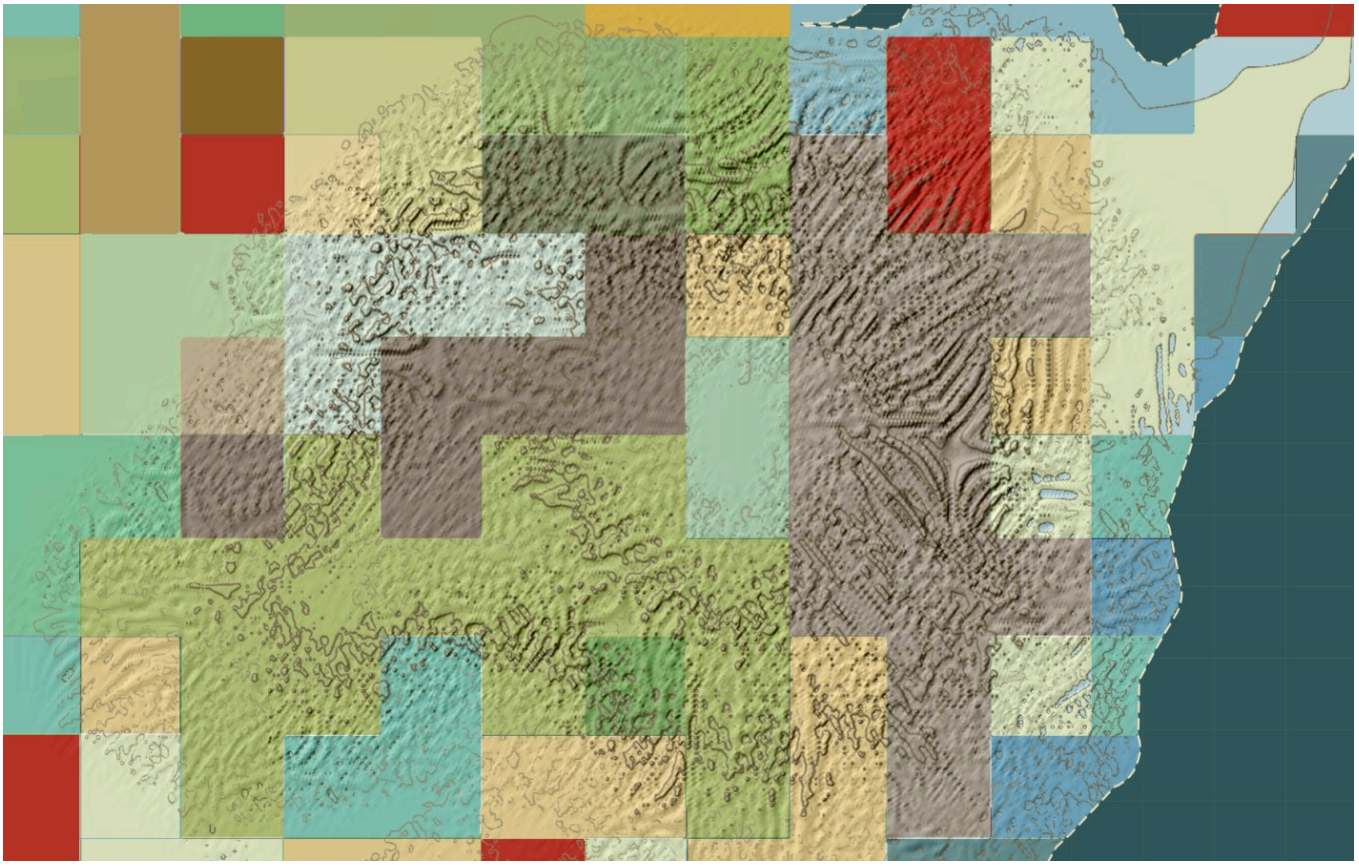


Figure 15. An earlier Codex build. Owner: "the elevation contours actaully look pretty good!! but everything else doesn't". The cover is still in parent-sized squares.

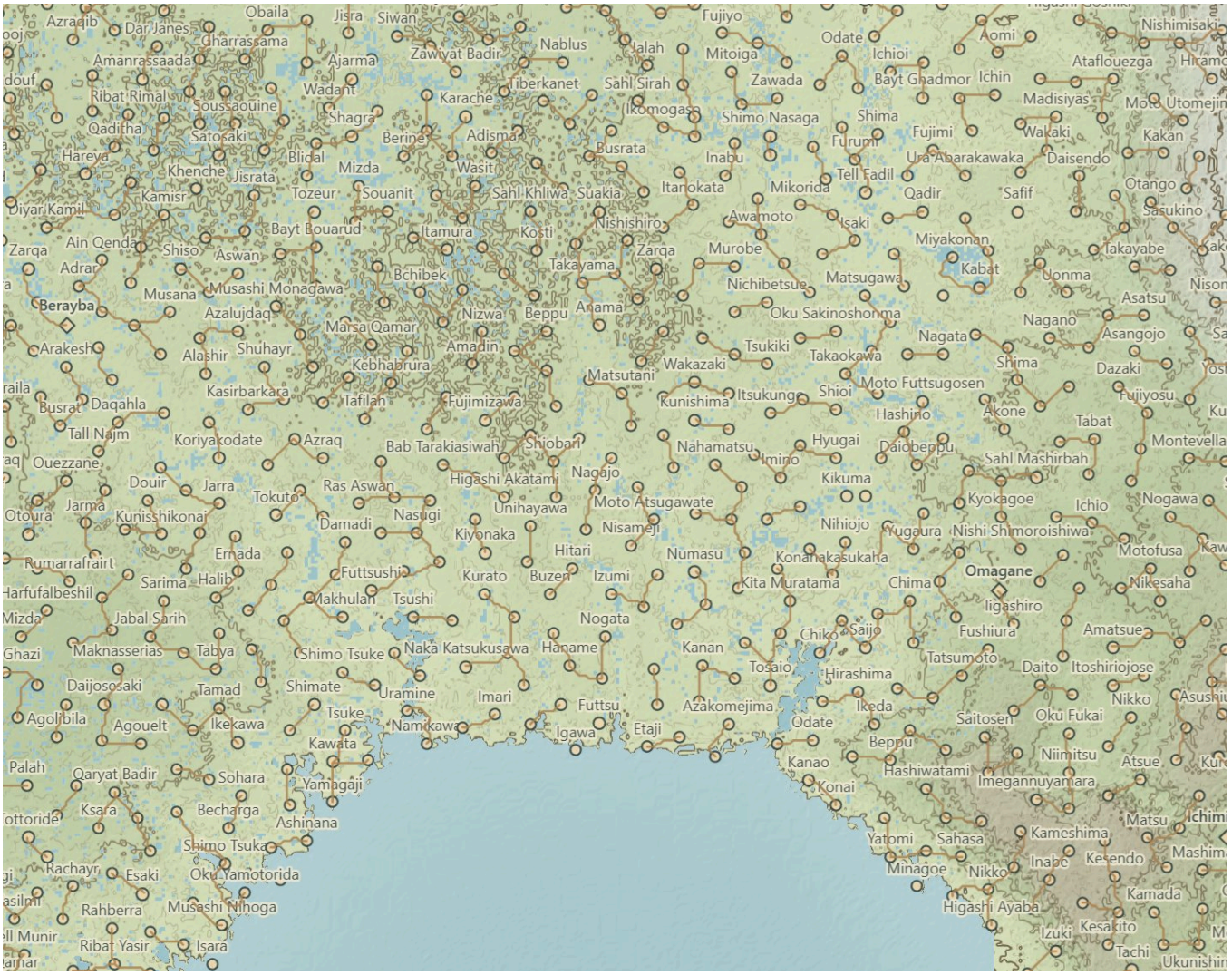


Figure 16. Owner: "the clutter is still insane"

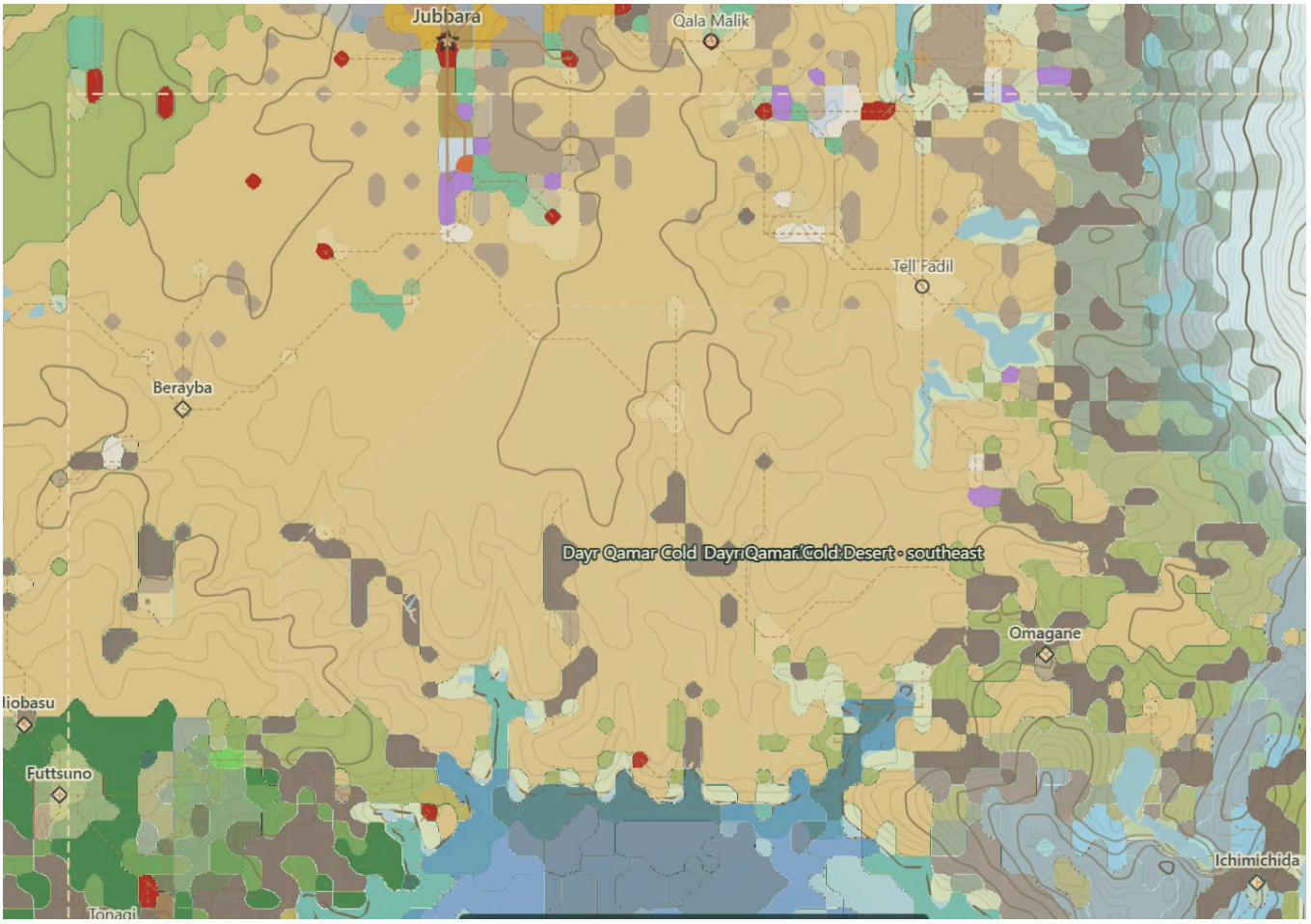


Figure 17. Owner: "the 'cold desert' became mostly a huge inalnd marsh … i would've expected the coast to become marshy, the south east to become mountainous, and the desert to remain desert."

1.4 The owner's question

> "are you even aware of what algorithms and techniques you should be using? does the generation not provide the correct data to work from?"

> "i think this needs a fundemtal change in approach, or, your latest change replaced some wrong and complex with something more accurate and simplisitc."

Part 2 — Investigation

This part describes how the code currently works. It names the conceptual errors behind both agents' attempts and answers whether the world generation supplies the data refinement needs.

2.1 Summary of findings

1. **Both agents treated refinement as a resolution problem, and the owner asked for a representation problem.** Each attempt asked "what does the ground look like at 2.4 km?" and then drew everything above a fixed threshold. The owner asked "what features matter on a map of this region?" A satellite-style answer is wrong for the product even when it is physically plausible. This is the owner's goal 1, and neither agent addressed it directly.

2. **The terrain method cannot create peaks.** The current method only cuts valleys into a smoothed copy of the parent surface. A parent elevation is an average over a ~14 km cell. Real summits stand well above that average and real valleys cut well below it. A method that only lowers the surface, and is pulled back to the averaged parent, can only produce one smooth dome per parent high with gullies on its flanks (Figures 3–4).
3. **River and lake rules use absolute thresholds, so density is uniform.** A stream is drawn wherever its catchment passes a fixed number of map samples. On a smooth dome, that creates evenly spaced parallel streams (Figures 3, 5). Lakes were first created by noise (Codex: too many, Figures 8–12). Claude then forbade them unless the parent had one (none in headwaters, Figure 3). Neither rule asks *why* a lake would be there.
4. **The world generation already holds most of the causes, but refinement ignores them.** Glacial erosion history, tectonic strain, landform classes, world-published peaks, ranges and cirques, and a relative-significance model for publishing features all exist at world scale. Regional terrain reads none of them.
5. **Some needed data is missing.** The world does not record the structural grain of a mountain range (its strike direction and how it is made: fold belt, volcanic arc, fault block). It does not record how rugged the ground is inside a cell. And it does not give features a rank that persists across map levels.
6. **Settlements:** inherited towns keep their exact parent coordinates, which were chosen on a 14 km grid. On a 2.4 km map those points can fall on steep slopes. New villages are placed by a fixed spacing rule, which produces a lattice (Figure 7). Both are verified in code. Which of the cliff-side towns in Figure 3 are inherited and which are new has not been checked.
7. **Process failure:** both agents judged success by counts, timings and unit tests (no pits, fewer rivers, fast enough). Those numbers could pass while the map still looked wrong. The owner's "too fixated on the numbers" is accurate.

2.2 How refinement works today (`bcca3a6`)

The parent world's raster fields are sampled onto a regular grid for the region: elevation, rock, fracture, glacial erosion, tectonic strain, ice, temperature, rainfall, runoff, soil, cover classes, flow, river order, population, magic and others. Parent features (rivers, towns, peaks) come along as vectors. The stages then run in this order:

Stage	What it does now
Terrain	Blurs the parent surface to remove its patch seams. Adds small fractal roughness, scaled to 0.3 × the local parent relief. Runs 48 steps of a river-incision plus hillslope-smoothing model that is pulled back toward that target every step. Fills every closed hollow except where the parent had a lake.
Water	Routes flow downhill. Marks a river wherever the catchment exceeds a fixed size (128 map samples, about 740 km² at the first level, scaled by local runoff). Rejects sheet flow and very short headwaters.
Geography	Turns river cells into line features: one per stream, from its source to the river it joins.
Landforms	Classifies ridge, valley, peak and pass from the new surface. Publishes peaks by prominence.
Cover	Keeps the parent's cover class, with the class edges warped smoothly. Changes it only for cold new peaks, floodplains, coastal flats, steep bare ground and lakes.
Places	Keeps inherited towns at their parent coordinates. Adds villages where population supports them: best site class first, with a minimum spacing between villages. Connects them with least-cost paths.

Codex's earlier pipeline (Figures 6–17) differed mainly in two ways. Its terrain was "interpolated parent + noise + local incision". It derived water, lakes and cover from thresholds on that noisy surface, so every noise hollow became a lake and every noise channel became a river.

2.3 Symptom by symptom

"One giant ridgeline", no individual peaks (Figures 3, 4, 6). The parent stores one elevation per ~14 km cell, which is in effect a mean surface. A mountain range at that scale is a smooth bump a few cells wide. Both pipelines treat that bump as the shape to keep and only add small perturbations to it.

- Codex added noise, so the bump became noisier (Figure 6).
- Claude added incision. Incision only removes material, and the restoring pull keeps the surface near the averaged bump, so the bump gained gullies (Figure 4).

Neither can raise a summit 500–1,500 m above the parent mean, as real massifs stand above their range's average. Neither places cols between summits, or longitudinal valleys that split a range into sub-chains. A uniform uplift field with uniform erodibility on a smooth dome is also the textbook setup for *consequent* drainage: streams run straight down the slope, roughly parallel or radiating. That is exactly what Figures 3–5 show.

"Rivers are way too even" (Figures 3, 5) and earlier "a shitload of rivers" (Figures 9, 13). Both pipelines decide whether to draw a stream with an absolute catchment threshold, measured in samples of the current map.

- Codex's threshold was low, so every sliver of flow was drawn.
- Claude's threshold is higher and the same at every level, so a map draws every stream above one size.

On uniform terrain under uniform rain, catchments reach that size at regular intervals, so the drawn streams are evenly spaced. Real valley spacing is in fact fairly regular at a given scale (Perron et al. 2009, in Part 3). Real maps do not look regular because cartographers *select*: the large rivers, then the ones that matter locally, with density varying by climate and rock. Neither pipeline selects. There is also no carry-over of identity: a world river does not keep its name, and the regional map has no notion of "this is the Verde, the next tier down from the Salt".

Lakes: too many (Codex), then none in the headwaters (Claude). Codex's lakes were hollows in the noise (Figures 8, 11, 12). Claude's rule removes every hollow unless the parent had a lake there. Both are size or provenance rules, and neither is a cause. The owner's rock barren expectation is a few larger lakes at river sources (Figure 10). That is what glaciated mountain terrain has: cirque tarns, chains of lakes in glacial valleys, lakes behind moraines. The world generation records cumulative glacial erosion and ice thickness (`glacialErosion`, `iceThickness`). Refinement samples both fields and uses neither when deciding where lakes form.

Cities on steep cliff sides (Figure 3). Inherited towns are copied to their exact parent coordinates and never re-sited. Those coordinates came from a 14 km grid. On a 2.4 km map they can land on a valley wall. The owner has said the child need not match the parent, so moving a town to the best site within its parent cell is allowed. Newly added villages reject slopes above 0.12 (about 7°), but the slope is measured per 2.4 km sample, which smooths out cliffs.

Towns evenly spaced (Figure 7) and "cities are better ish" (Figure 3). New villages are chosen best-site-first, then excluded within a fixed spacing of any existing seat. With a uniform spacing rule, the result tends toward a lattice. Real settlement patterns are clustered along valleys, coasts, river crossings and confluences, and are sparse elsewhere.

Cover: patches, holes and the cold-desert marsh (Figures 13–17); "keep the parent class" rejected by the owner. Codex re-derived cover from thresholds on noisy refined fields, which produced patches and the inland marsh. Claude's fix kept each sample's parent class. That removes the errors but also removes the detail. A world-scale "forest" cell is a summary of a mosaic: forest with river meadows, rock outcrops and alpine ground above the

tree line. The refined map should reveal that mosaic, driven by the new terrain: elevation zones, valley floors, slope direction and drainage. It should not repaint the summary label.

2.4 Does the world generation provide the right data?

Mostly yes, but refinement uses little of it.

World data	Present at world scale	Used by refinement terrain/water/lakes
Elevation	yes	yes (as a surface to keep)
Rock type, fracture	yes	yes (erodibility only)
Rainfall, runoff, temperature	yes	yes
Tectonic strain	yes	**no**
Cumulative glacial erosion, ice thickness	yes	**no** (only published or used for magic)
Landform class (range, plateau, valley …)	yes	**no** (re-classified from the new surface)
Published world peaks, ranges, cirques, fjords	yes	**no**
Published world rivers (named, with discharge)	yes	only as discharge entering at the frame edge; names are not carried
Relative feature significance (e.g. landscape share of the world extent)	yes, for world publication (`worldSignificance.ts`)	**no** — regional uses absolute thresholds
Range structure: strike direction, range type (fold belt, volcanic arc, fault block), main divide	**not recorded**	—
Ruggedness inside a cell (relief within a world cell)	**not recorded**	—
Feature rank that persists across map levels	**not recorded**	—

The missing items are the ones that would let a regional map lay out *distinct* peaks, divides and valleys in the right orientation. The unused items are the ones that would give lakes and peaks their causes.

2.5 Why the iterations kept missing

- **The target was implicit.** No written description of a correct regional map existed, so each iteration fixed the last complaint: too many lakes became no lakes, too many rivers became fewer evenly spaced rivers.
- **Validation by numbers.** Tests asserted no pits, bounded counts and timings. None asserted what the owner judges: distinct summits, varied drainage density, lakes with causes, settlements at sites.
- **Simulation as the answer to goal 1.** Physical simulation gives authenticity (goal 2). It does not decide what to *show* (goal 1). Both agents expected a better simulation to produce a better map. Cartographic selection is a separate step with its own rules.

Part 3 — Research: techniques that fit this problem

3.1 Terrain amplification: adding real structure under a coarse surface

This is a well-studied problem in computer graphics under the name *terrain amplification*. A useful survey is Galin et al., "A Review of Digital Terrain Modeling", *Computer Graphics Forum* 38(2), 2019 (Eurographics STAR). The main families:

- **Uplift + stream-power erosion.** Cordonnier, Braun, Cani, Benes, Galin, Peytavie and Guérin, "Large Scale Terrain Generation from Tectonic Uplift and Fluvial Erosion", *CGF* 35(2), 2016. The input is an *uplift map*, not a height map. Erosion acting against uplift produces ridges, summits, cols and dendritic valleys, and the elevations are an output. The key difference from the current code: here the coarse map controls *uplift*, and the heights, including summits above the mean, are produced by the process. The current code holds heights to the coarse surface, which prevents that.
- **Multi-scale erosion amplification.** Schott, Galin, Guérin, Peytavie and Paris, "Terrain Amplification using Multi-scale Erosion", *ACM TOG* 43(4), 2024 (SIGGRAPH). This turns a low-resolution terrain into a high-resolution one that stays hydrologically consistent. It runs fast approximations of thermal, stream-power and deposition processes at successive scales. This is the closest published match to "refine a region from its parent".
- **Orometry: peaks and saddles as the skeleton.** Argudo, Galin, Peytavie, Paris, Gain and Guérin, "Orometry-based Terrain Analysis and Synthesis", *ACM TOG* 38(6), 2019 (SIGGRAPH Asia). From a coarse elevation map, it first builds a *graph of peaks and saddles* whose prominence and spacing follow statistics measured on real mountain ranges. It then fills in a continuous surface with a river network. This directly answers "major peaks should start to appear": peaks are generated as features with prominence, not hoped for from noise.
- **Hydrology first.** Génevaux, Galin, Guérin, Peytavie and Beneš, "Terrain Generation Using Procedural Models Based on Hydrology", *ACM TOG* 32(4), 2013. It generates the river network first, with Horton–Strahler ordering and watersheds, and builds terrain around it. Cartographic river selection then follows from the river hierarchy itself.
- **Procedural dendritic patterns.** Gaillard et al., "Dendry: A Procedural Model for Dendritic Patterns", I3D 2019. It evaluates branching valley patterns locally from a seed. That is useful when regions must be generated independently and stay deterministic (R6).

3.2 Geomorphology that should drive the parameters

- **Valley spacing.** Perron, Kirchner and Dietrich, "Formation of evenly spaced ridges and valleys", *Nature* 460, 2009. At a given scale, the competition between channel incision and soil creep fixes a characteristic valley spacing. So regular spacing *within one tier* is realistic. What makes real maps look irregular is the hierarchy of tiers and the cartographer's selection, not randomness in the ground.
- **Range structure.** Real ranges are not one dome. A fold-and-thrust belt has parallel ridges and long valleys along its strike (trellis drainage), cut across by gorges. A volcanic arc has discrete cones. A fault-block range has one steep front and one gentle back slope. This structure follows the tectonic setting, which the world simulation knows (plate motion, strain, volcanism) but does not hand to refinement.
- **Lakes have causes.** In mountains these are mainly glacial:
  - cirque tarns behind a rock lip or till dam;
  - chains of lakes down stepped glacial valleys;
  - long lakes in over-deepened troughs behind moraines. Other causes are tectonic basins (grabens), landslide dams, karst and volcanic craters. Lake *density* is high in recently glaciated hard rock and near zero in unglaciated dry ranges. That density can be read from the world's glacial-erosion, ice-history and rock fields.

3.3 Cartographic generalization: what to show at each level

This is the core of goal 1. Cartographers have formal methods for it.

- **Principle of Selection (Radical Law).** Töpfer and Pillewizer, "The Principles of Selection", *The Cartographic Journal*, 1966. The number of features shown scales with the square root of the scale ratio. For the same piece of ground, a child map at about 6× finer scale shows roughly $\sqrt{6} \approx 2.4\times$ as many features as its parent does. That is far fewer than the 36× more raster samples it has. The law gives *how many* to keep, not *which*.
- **Selection by upstream drainage area, stratified by local density.** Stanislawski, "Feature pruning by upstream drainage area to support automated generalization of the United States National Hydrography Dataset", *Computers, Environment and Urban Systems* 33(5), 2009. The USGS prunes rivers by drainage area. It does this separately in each sub-basin, with a target density for each, so wet dissected country stays denser than dry country *after* generalization. That is the direct fix for "rivers way too even".
- **Rank attributes for zoom.** Natural Earth publishes `scalerank` and `min_zoom` on rivers and lakes: a curated rank of how important each feature is regionally. A renderer shows features down to the rank that fits the zoom. The atlas needs an equivalent: each feature carries a rank that is stable across map levels. The world map shows ranks 1–2; a region shows ranks 1–4, and so on. The Mississippi/Verde/Lynx Lake example from the owner is exactly this.
- **Exaggeration and simplification.** Cartographic relief traditions (e.g. Imhof, *Cartographic Relief Presentation*) emphasise main ridgelines, named peaks and passes, and suppress the rest. A peak earns a symbol and a name by prominence relative to the map's extent, not by being any local maximum.

3.4 Settlement siting

Standard settlement geography distinguishes the *site* from the *situation*:

- **Site factors:** a reliable water source (wet point); dry ground in marsh or floodplain (dry point); a ford or bridging point; a defensible hill or river bend; shelter; flat land.
- **Situation factors:** river confluences, valley junctions and gaps through ranges where routes meet; the lowest bridging point of a river; coastal harbours.

A refined map should first find these *sites* as features of the new terrain (confluences, valley mouths, fords, passes, lake ends, harbours). It should then fill them in order of rank, and thin them by hierarchy (a market town's hinterland), not by a fixed spacing. Inherited towns move to the best such site within their parent cell.

3.5 Putting it together: a candidate architecture

This is a recommendation for discussion. It is not a decided design.

1. **The parent supplies a specification of causes, not a surface to keep.** For each region, the parent provides uplift or relief intent (from elevation and tectonic strain), range type and grain, rock, climate, glacial history, named features with their ranks, and boundary conditions: where rivers, ridges and roads cross the region's edge. Elevation is a hint, never a constraint.
2. **Skeleton first.** Place the next tier of named features as structure before any raster:
    - peaks and saddles with prominence following an orometric distribution, laid out along the range's grain;
    - the main divide and sub-chains;

    - valley trunks that connect to the inherited rivers and their edge crossings. *(Orometry, hydrology-first methods.)*
3. **Physics fills between the skeleton.** Uplift from the skeleton plus stream-power and hillslope processes produce the continuous surface. Summits come out above the parent mean and valleys below it. *(Uplift + erosion; multi-scale amplification.)*
4. **Causal features.** Lakes are placed where a cause exists: glacial overdeepening and cirques where glacial erosion was high, tectonic basins, and dams. Cover reveals the mosaic inside each parent class, driven by elevation zones, valley floors, slope and drainage.
5. **Cartographic selection.** Every feature gets a rank. Each map draws only the tiers that fit its scale, with counts guided by the Radical Law. Density targets are set per sub-basin, so dry country stays sparse. Inherited features keep their names and ranks; new ones get the next ranks.
6. **Independence for collaboration.** A region's output depends only on the parent specification, its seed and its boundary conditions. Two neighbouring regions agree at the edges because they share the parent's crossing points, not because either reads the other.

Expected cost: the skeleton and selection steps are graph operations on hundreds to thousands of features and are cheap. The physics step can reuse the existing solver, run with uplift as its input instead of being pulled back to a target. Region creation and boundary finding are untouched.

3.6 Open questions for the owner

1. **Rank across levels.** Should a feature's rank be assigned once, when it is first created, and inherited by every deeper level? Or re-judged on each map relative to that map's extent? The Natural Earth practice is a stable rank.
2. **Data additions.** Is adding range type and structural grain to the world output acceptable? These would come from tectonic history the simulation already runs.
3. **Exaggeration.** May the regional map exaggerate relief for legibility, for example by making the main peaks more prominent than physics alone would?
4. **Acceptance.** The owner's judgement of rendered maps has been the real test. A short written checklist would give future iterations a target that is not a count. For example: distinct named summits on a range; drainage denser in wet country than dry; lakes only with a visible cause; settlements at confluences, crossings and coasts.

Sources

- Galin et al. 2019, "A Review of Digital Terrain Modeling", CGF 38(2) — https://diglib.eg.org/handle/10.1111/cgf13657 ; PDF https://www.cs.purdue.edu/homes/bbenes/papers/Galin19CGF.pdf
- Cordonnier et al. 2016, "Large Scale Terrain Generation from Tectonic Uplift and Fluvial Erosion", CGF 35(2) — https://diglib.eg.org/handle/10.1111/cgf12820 ; PDF https://www.cs.purdue.edu/homes/bbenes/papers/Cordonier16CGF.pdf
- Schott et al. 2024, "Terrain Amplification using Multi-scale Erosion", ACM TOG 43(4) — https://h-schott.github.io/publications/mserosion/publi_mserosion.html
- Argudo et al. 2019, "Orometry-based Terrain Analysis and Synthesis", ACM TOG 38(6) — https://pubs.cs.uct.ac.za/id/eprint/1342/

- Génevaux et al. 2013, "Terrain Generation Using Procedural Models Based on Hydrology", ACM TOG 32(4) — https://www.cs.purdue.edu/homes/bbenes/papers/Genevaux13ToG.pdf
- Gaillard et al. 2019, "Dendry: A Procedural Model for Dendritic Patterns", I3D — https://cs.purdue.edu/cgvlab/www/publications/Gaillard19I3D
- Perron, Kirchner & Dietrich 2009, "Formation of evenly spaced ridges and valleys", Nature 460 — https://dspace.mit.edu/handle/1721.1/64681
- Töpfer & Pillewizer 1966, "The Principles of Selection" (Radical Law); discussion in Jiang, https://arxiv.org/pdf/1102.1561
- Stanislawski 2009, "Feature pruning by upstream drainage area …", CEUS 33(5) — https://www.usgs.gov/publications/feature-pruning-upstream-drainage-area-support-automated-generalization-united-states ; stratified density comparison https://kartographie.geo.tu-dresden.de/downloads/ica-gen/workshop2011/genemr2011_Stanislawski.pdf
- Natural Earth river/lake rank attributes (`scalerank`, `min_zoom`) — https://deepwiki.com/elastic/natural-earth-vector/4.3-physical-features
- Glacial lakes and overdeepening — https://en.wikipedia.org/wiki/Overdeepening ; https://en.wikipedia.org/wiki/Glacial_landform
- Settlement site and situation — https://geographyfieldwork.com/SiteSituation.htm